

# RealityCheck: An Evidence-Grounded Hallucination Correction Pipeline for Large Language Models

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## Abstract

Large language models (LLMs) have demonstrated remarkable capability in generating fluent, contextually coherent responses across diverse domains; however, they remain susceptible to hallucination — the generation of factually incorrect or unverifiable content presented with apparent confidence. This tendency poses significant risks in high-stakes applications such as healthcare, legal practice, and academic research, where the consequences of misinformation can be severe. Existing approaches to this problem, including retrieval-augmented generation and post-hoc detection frameworks, address hallucination at the generation stage or diagnose it after the fact, but do not provide a mechanism for inline correction of erroneous claims before the response reaches the user.

We present RealityCheck, a modular, model-agnostic, six-phase hallucination correction pipeline that operates as an external verification layer over existing LLMs without modifying their internal parameters. The system extracts factual claims from LLM-generated responses, retrieves Wikipedia-grounded evidence through a cache-first retrieval architecture, verifies each claim through a multi-signal framework combining Natural Language Inference, semantic alignment scoring, and rule-based reasoning, and produces a corrected response through deterministic answer synthesis. Evaluated on the TruthfulQA benchmark

across IBM WatsonX Granite, Meta Llama, and MistralAI Mistral, RealityCheck achieves substantial improvements in claim-level accuracy and hallucination recall across all three models, while maintaining low overcorrection rates. The results demonstrate the potential of evidence-grounded, claim-level correction pipelines for improving the factual reliability of LLM-generated content in high-stakes deployments.

**Keywords:** Large Language Models, Hallucination Mitigation, Fact Verification, Retrieval-Augmented Verification, Natural Language Inference, Evidence-Grounded AI, Answer Synthesis, Trustworthy AI

## 1 Introduction

Large Language Models (LLMs) such as ChatGPT, Claude, Gemini, and Grok have witnessed rapid and widespread adoption across a broad spectrum of domains, including healthcare, legal practice, academic research, and education. Their appeal lies in their capacity to generate fluent, contextually coherent, and professionally articulated responses with minimal user effort, effectively positioning them as accessible knowledge interfaces for both expert and general audiences.

However, this growing reliance on LLMs has brought a critical reliability concern to the forefront, a phenomenon widely referred to as hallucination, wherein a model generates outputs that are factually incorrect, internally inconsistent, or entirely insufficiently verifiable, yet presents them with a tone of authority and confidence. This characteristic is particularly concerning because the linguistic fluency and structural coherence of LLM-generated text can create a strong impression of credibility, discouraging users from independently verifying the information against primary sources.

The implications of unchecked hallucination extend well beyond mere inconvenience. In healthcare, the propagation of false medical claims can lead to dangerous misinformation that undermines clinical decision-making. In academic research, fabricated citations or erroneous statistical claims erode the integrity of scholarly discourse and introduce false benchmarks into the literature. In public-facing applications, hallucinated content risks spreading misinformation at scale, influencing public perception on the basis of non-existent evidence. In each of these contexts, the consequences of misplaced trust in LLM-generated content can be severe and, in some cases, irreversible.

Addressing this problem has been an active area of research. Existing approaches include Retrieval-Augmented Generation (RAG) [1], Natural Language Inference (NLI)-based verification [2], and knowledge-grounded generation frameworks [3]. While these methods have demonstrated promise, they are not without limitations. RAG-based systems retrieve relevant context but do not always guarantee that retrieved information is used to correct erroneous claims already present in the generated output. NLI-based approaches, while effective at entailment classification, often struggle with compound reasoning and multi-hop contradiction resolution. Furthermore, many existing systems are designed to detect hallucinations as a post-hoc

diagnostic rather than as an inline correction mechanism that refines the response before it reaches the user.

In this paper, we propose a six-stage hallucination mitigation pipeline that addresses this gap by performing inline correction, factual claim extraction, Wikipedia-grounded verification, and response refinement before delivering a corrected answer to the user. Unlike conventional hallucination detection systems that operate purely as post-hoc evaluators, the proposed framework functions as an inline corrective architecture that actively reconstructs the final response using evidence-backed claim synthesis. The pipeline is orchestrated through a Live Console module that allows flexible selection of the underlying LLM, currently supporting IBM WatsonX, Meta’s Llama, and MistralAI, enabling controlled, model-specific evaluation. Factual claims are extracted from LLM responses using sentence transformer based semantic extraction, verified against the evidence-grounded verification primarily driven by Wikipedia-based retrieval, and classified as supported, contradicted, or insufficiently verifiable. Contradicted claims are corrected or replaced using evidence-grounded synthesis mechanisms, while insufficiently verifiable claims are hedged with appropriate epistemic qualifiers to preserve informational value without asserting unconfirmed facts.

The key contributions of this work are as follows:

- A modular six-stage hallucination mitigation architecture integrating retrieval, contradiction-aware verification, and evidence-grounded response synthesis across multiple LLMs.
- A claim-level verification mechanism grounded in Wikipedia, with a three-way classification scheme: supported, contradicted, and insufficiently verifiable.
- An evidence-grounded answer synthesis framework capable of reconstructing corrected responses by preserving supported claims, correcting contradicted claims, and softening insufficiently verifiable assertions.
- Empirical evaluation on the TruthfulQA benchmark demonstrating the effectiveness of the proposed approach across IBM WatsonX, Meta Llama, and MistralAI.

The remainder of this paper is organized as follows. Section II discusses related work and existing hallucination mitigation approaches. Section III presents the proposed methodology and pipeline architecture. Section IV describes the experimental setup and evaluation strategy. Section V discusses the obtained results and limitations, while Section VI concludes the paper and outlines future research directions.

## 2 Related Work

### 2.1 Hallucination in Large Language Models

The problem of hallucination in natural language generation has been extensively studied in recent years. Ji et al. [3] provide a comprehensive taxonomy of hallucination in NLG systems, distinguishing between intrinsic hallucinations, where model outputs contradict the source context, and extrinsic hallucinations, where outputs introduce unverifiable or fabricated information not grounded in any available evidence. Their survey establishes that hallucination is not an incidental failure mode but a systemic

property of statistical language models optimised for fluency over factual fidelity. More recently, Farquhar et al. [2] demonstrated that semantic entropy derived from model output distributions can serve as a reliable signal for hallucination detection, offering a probabilistic framework for identifying unreliable claims. While these works deepen the theoretical understanding of hallucination, they do not provide inline correction mechanisms that refine responses before delivery to the user, a gap that motivates the present work.

In an effort to address hallucination through model self-deliberation, Dhuliawala et al. [4] proposed Chain-of-Verification (CoVe), a prompting framework in which a model drafts an initial response, generates targeted verification questions about its own claims, answers those questions independently to avoid self-reinforcing bias, and produces a final revised response. While CoVe demonstrates measurable reductions in hallucination across list-based and open-domain tasks, it relies entirely on the model’s internal parametric knowledge for both generation and verification, an approach that is inherently limited when the model itself lacks accurate knowledge of a domain. Unlike CoVe, our pipeline grounds verification in an external, authoritative knowledge source rather than the model’s own outputs.

## 2.2 Fact Verification and Knowledge-Grounded Systems

Fact verification against structured knowledge sources has been a long-standing challenge in NLP. Thorne et al. [5] introduced FEVER (Fact Extraction and VERification), a large-scale benchmark comprising 185,445 claims derived from Wikipedia, each labelled as *Supported*, *Refuted*, or *NotEnoughInfo*. This three-way classification scheme has become a foundational reference point for claim-level verification tasks, and the use of Wikipedia as a grounding source has remained prominent in subsequent work.

## 2.3 Response Correction via External Knowledge Grounding

Beyond detection, a smaller body of work has explored the active correction of hallucinated content in LLM outputs. Guan et al. [6] proposed Knowledge Graph-based Retrofitting (KGR), a framework that extracts factual statements from LLM-generated responses, validates them against a knowledge graph, and retrofits incorrect statements with verified alternatives. KGR demonstrates that post-generation correction is a viable and effective strategy, particularly for compound reasoning tasks. However, its dependence on structured knowledge graphs limits its applicability to domains where such graphs are available and sufficiently comprehensive. Our approach addresses this constraint by grounding verification in Wikipedia, a broad, continuously maintained, and domain-agnostic free-text corpus, thereby extending the correction capability to a substantially wider range of queries.

Min et al. [7] extended this paradigm to long-form text generation through FActScore, a metric that decomposes generated text into atomic facts and evaluates each independently against a knowledge source. While FActScore provides fine-grained factuality measurement, it is designed as a post-hoc evaluation framework rather than a pipeline-driven correction system, and does not modify or refine the original response.

## 2.4 Retrieval-Augmented and Wikipedia-Grounded Generation

Retrieval-Augmented Generation (RAG), introduced by Lewis et al. [1], represents a prominent approach to grounding language model outputs in external knowledge by conditioning generation on retrieved passages. While RAG demonstrably improves factual consistency in knowledge-intensive tasks, it operates at the generation stage, retrieved content informs what the model generates, but does not guarantee post-hoc correction of erroneous claims already present in the output. WikiChat [8] takes a closer step toward our objective by grounding LLM responses in live Wikipedia content. Its pipeline extracts claims from LLM-generated responses, verifies them individually against Wikipedia, and filters out unverified facts before composing the final response. However, WikiChat operates primarily as a conversational agent built on few-shot prompting of a single underlying model, and does not provide a modular, model-agnostic pipeline for inline hallucination correction across multiple LLMs. Furthermore, its handling of unverifiable claims, simply omitting them, results in potential information loss, a limitation our hedging strategy is designed to address.

More recent work has explored ensemble-based strategies for hallucination mitigation, wherein outputs from multiple models are compared for consistency rather than verified against a fixed external source. Dey et al. [9] proposed Uncertainty-Aware Fusion (UAF), which combines responses from multiple LLMs weighted by their estimated confidence and factual accuracy, achieving improvements over single-model baselines on factoid question-answering benchmarks. While ensemble methods such as UAF benefit from the diversity of model knowledge, they remain bounded by the collective parametric knowledge of the participating models and offer no mechanism for verifying claims against ground-truth external sources. Our pipeline complements this line of work by introducing explicit external grounding as the verification signal, independent of model confidence.

## 2.5 NLI-Based Verification Approaches

Natural Language Inference (NLI) models, which classify the relationship between a premise and a hypothesis as entailment, contradiction, or neutral, have been widely adopted for hallucination detection [2]. NLI-based systems assess whether generated claims are logically supported by a reference source, providing a structured framework for consistency verification. However, their effectiveness is contingent on the quality of claim decomposition and evidence retrieval, and they often struggle with compound or multi-hop reasoning chains where a single claim may depend on multiple facts drawn from disparate sources [3]. These limitations motivate the use of semantic similarity-based decomposition in our pipeline.

## 2.6 Sentence Transformers for Semantic Decomposition

Sentence-level semantic representations have become a standard tool in NLP pipelines requiring fine-grained text analysis.

Reimers and Gurevych [10] introduced Sentence-BERT (SBERT), a modification of the BERT architecture employing siamese and triplet network structures

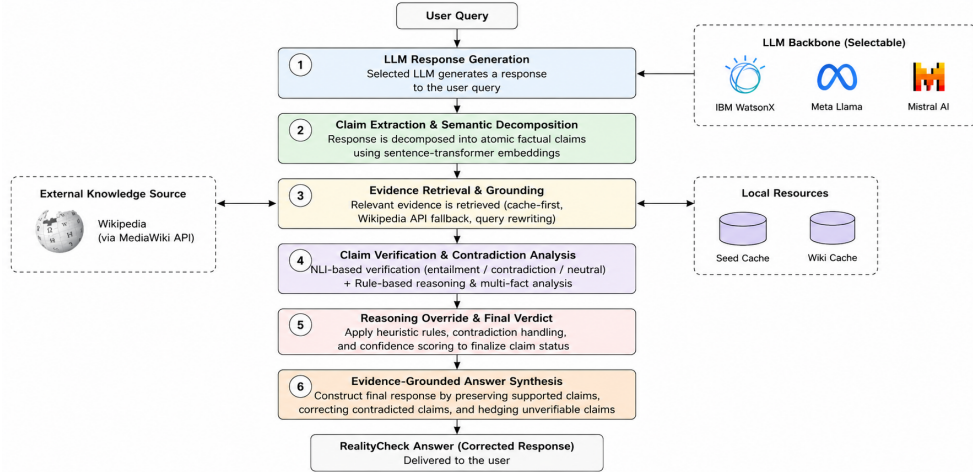
to produce semantically meaningful sentence embeddings efficiently comparable via cosine similarity. The `multi-qa-MiniLM-L6-dot-v1` model, a lightweight variant from the sentence-transformers family, has demonstrated strong performance on semantic retrieval tasks. In the present work, this model is employed for semantic decomposition of LLM responses into discrete factual units suitable for downstream verification. Sentence-level semantic embeddings are employed as an enabling mechanism for claim decomposition within the proposed pipeline.

The work reviewed above reflects substantial progress across complementary dimensions of the hallucination problem, from theoretical taxonomies [3] and probabilistic detection [2], to retrieval-augmented generation [1], Wikipedia-grounded chatbots [8], self-verification through model deliberation [4], and external knowledge-graph correction [6]. Yet a unifying limitation persists across these approaches: none simultaneously addresses detection, external grounding, inline correction, and epistemic hedging of unverifiable claims within a single, model-agnostic pipeline. RAG-based systems intervene at generation time but cannot retroactively correct erroneous claims. Self-verification methods such as CoVe remain constrained by the boundaries of parametric model knowledge. Knowledge graph approaches offer structured correction but require curated, domain-specific resources unavailable for general-purpose use. Wikipedia-grounded systems like WikiChat filter unverified claims entirely, sacrificing informational completeness. The proposed system addresses these gaps holistically by grounding verification in a broad, domain-agnostic knowledge source while performing inline correction of contradicted claims. Additionally, the framework preserves informational utility through epistemically hedged representations of insufficiently verifiable claims, thereby offering a more complete and practically deployable hallucination mitigation pipeline than prior isolated approaches.

## 3 Proposed Methodology

### 3.1 System Overview

RealityCheck is a modular, evidence-grounded, six-phase hallucination correction pipeline designed to intercept factually incorrect content in LLM-generated responses before delivery to the user. The system receives a user query, forwards it to a user-selected large language model, extracts the factual claims embedded in the generated response, retrieves externally grounded evidence, verifies each claim individually, and synthesises a corrected response in which contradicted claims are repaired and unverifiable claims are epistemically hedged. The complete architecture of the pipeline is illustrated in Figure 1. The design of RealityCheck is intentionally model-agnostic and pipeline-driven: each phase operates as an independent, replaceable module with well-defined inputs and outputs, and the system currently supports three LLMs, IBM WatsonX Granite, Meta Llama, and MistralAI Mistral, without any modification to the verification or synthesis logic.



**Fig. 1** Full architecture of the RealityCheck six-phase hallucination correction pipeline.

## 3.2 Dataset

The TruthfulQA benchmark [11] is used as the evaluation corpus for this work. TruthfulQA comprises 817 questions spanning 38 categories, including health, law, finance, history, and popular misconceptions, specifically designed to elicit imitative falsehoods from language models that have learned to reproduce commonly held but factually incorrect beliefs. The dataset provides, for each question, a best answer, a set of correct answers, and a set of incorrect answers, as well as multiple-choice target structures with per-choice binary labels. Prior to evaluation, the raw dataset is preprocessed to normalise whitespace, parse structured answer fields, and merge the generation and multiple-choice validation splits into a unified record format with stable question identifiers. This normalised representation serves as the input corpus for all downstream experimental phases.

## 3.3 LLM Response Generation

A stratified random sample of questions is drawn from the preprocessed TruthfulQA corpus with a fixed random seed to ensure reproducibility across experimental runs. Each sampled question is submitted to the selected LLM using a structured two-part prompt. The system prompt instructs the model to behave as a factual question-answering assistant, constrains the response to a maximum of three sentences, prohibits question restatement, meta-commentary, and unsolicited disclaimers, and directs the model to acknowledge rather than confabulate answers to subjective or time-sensitive questions. The user prompt enforces the delivery of a direct final answer without preamble or structural formatting. The raw response is subsequently cleaned to remove markdown artefacts and formatting noise, and a response quality assessment is applied to flag outputs that are empty, excessively short, or evasive. The

complete generation record, including the cleaned response, quality flags, and raw API output, is retained as the input to Phase 3.

### 3.4 Factual Claim Extraction

The cleaned LLM response is decomposed into discrete factual claims through a rule-based linguistic pipeline. Sentence segmentation is performed using punctuation-aware splitting, and each resulting sentence is assessed for *checkability*, the degree to which it constitutes a verifiable factual assertion. A sentence is classified as checkable if it is non-subjective, free of explicit hedging or opinion markers, and contains at least one linguistically motivated factual cue: a copular or auxiliary verb, a named entity pattern, a four-digit year reference, or a domain-relevant keyword. Sentences containing explicit subjectivity markers, such as "*opinions vary*" or "*no single factual answer*", are classified as non-checkable and excluded from downstream verification at this stage. For checkable claims, the module additionally detects context-dependent fragments, short sentences beginning with connective adverbs or referential pronouns, and enriches their evidence retrieval query by prepending the original question and the preceding sentence, thereby resolving anaphoric references before evidence retrieval. Each claim is assigned an importance label (*essential* or *supporting*) based on its sentence position, lexical significance cues, and total claim count in the response. The output of this phase is a structured record enumerating all extracted claims with their checkability status, normalised text, importance labels, and enriched evidence queries.

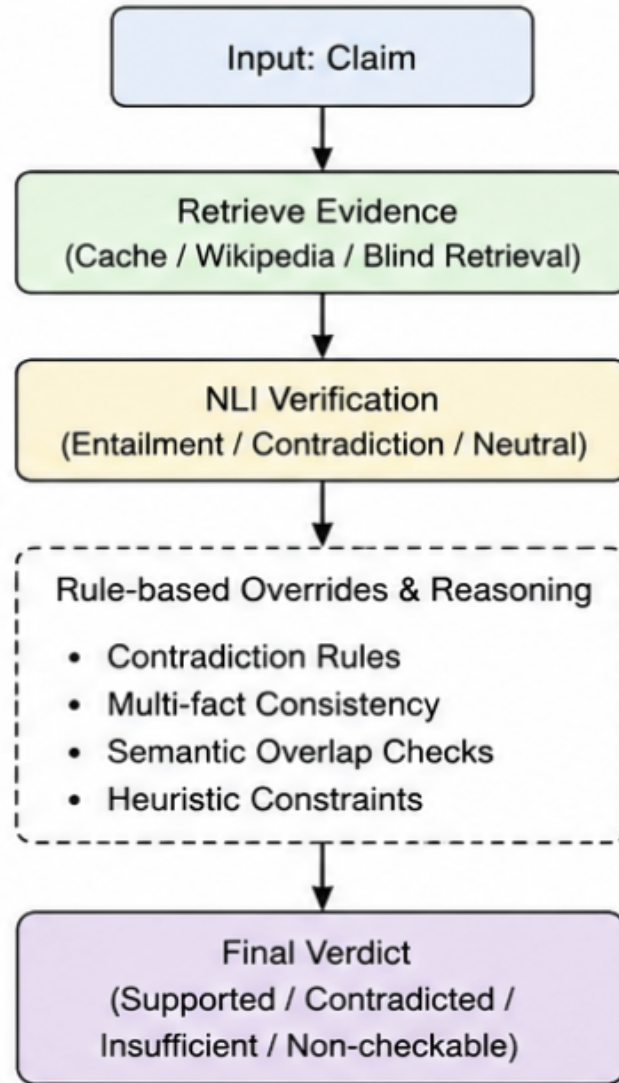
### 3.5 Wikipedia-Grounded Evidence Retrieval

For each checkable claim, relevant evidence is retrieved from Wikipedia through a cache-first retrieval architecture. The retrieval strategy operates across three priority layers. In the first layer, a local seed evidence bank, pre-populated with curated Wikipedia content for common TruthfulQA question categories, is consulted. If this bank returns sufficient evidence pages for a given claim, network retrieval is bypassed entirely, minimising API load and experimental latency. In the second layer, if the local seed is insufficient, the Wikipedia MediaWiki Action API is queried using the claim’s enriched evidence query, with an OpenSearch fallback triggered if the primary search yields no results. In the third layer, where a TruthfulQA question record carries a direct Wikipedia source URL, that page is fetched and treated as a high-priority evidence source. All retrieved content is cached to disk, ensuring that repeated experimental runs do not re-query the API. Retrieved page text is segmented into fixed-size overlapping chunks, each of which is encoded into a dense semantic representation using a sentence transformer model. Chunks are then ranked by semantic similarity against the claim text, and the top-k highest-scoring chunks are retained as the evidence set for that claim. This claim-level retrieval design ensures that verification in the subsequent phase operates on targeted, claim-specific evidence rather than on full-document content.



### 3.6 Claim Verification

Each checkable claim is verified against its retrieved evidence set through a multi-signal verification framework. The verification decision is produced by fusing four complementary signals rather than relying on any single classifier. The claim verification flow is illustrated in Figure 2. The first signal is *semantic similarity*, which measures the



**Fig. 2** Claim-level verification flow in Phase 5 of the RealityCheck pipeline.

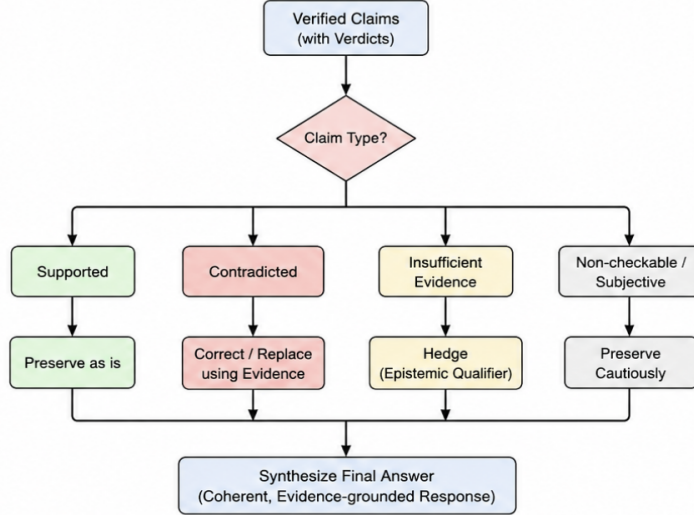
degree of topical and entity overlap between a claim and an evidence chunk, providing a relevance gate that prevents irrelevant evidence from influencing the verification decision. The second signal is *Natural Language Inference (NLI)*, applied at the chunk level using a cross-encoder NLI model to classify the relationship between the evidence chunk (as premise) and the claim (as hypothesis) as entailment, contradiction, or neutral. The third signal is a set of *rule-based support and contradiction overrides*, high-precision pattern-matching rules tuned to common TruthfulQA categories such as misattributed quotations, historical misconceptions, and compound multi-entity factual claims, which correct for known failure modes of NLI models on paraphrased or negated factual assertions. The fourth signal is *alignment score aggregation*, which applies a Noisy-OR aggregation function across multiple evidence chunks so that several medium-strength pieces of evidence can jointly support a claim that no single chunk alone would confirm. A dedicated temporal verification sub-module handles age and date-computation claims separately, resolving them through deterministic arithmetic over birth date evidence rather than through NLI classification, which is known to perform poorly on such claims. Each claim is assigned one of four verification labels: *supported*, *contradicted*, *insufficient evidence*, or *skipped (non-checkable)*, along with the best supporting and contradicting evidence chunks and a structured verdict reason.

### 3.7 Answer Synthesis

The verification outputs are consumed by a deterministic answer synthesis module that constructs the final corrected response without invoking any LLM. This deterministic design prevents secondary hallucination during the correction stage and ensures reproducibility of the synthesized response. The synthesis logic applies one of three actions to each claim based on its verification label. Claims labelled *supported* are retained as-is in the corrected response. Claims labelled *contradicted* are *repaired*, replaced with a corrected statement derived from the best contradicting evidence through a set of domain-aware correction templates for common TruthfulQA misconception categories. Claims labelled *insufficient evidence* are *omitted or softened*, either removed from the response or replaced with an epistemically hedged formulation that preserves informational value without asserting an unconfirmed fact. Non-checkable claims are passed through unchanged. The answer synthesis logic is illustrated in Figure 3. The corrected sentences are deduplicated, joined into a coherent response string, and labelled with one of six answer status categories: *fully supported*, *corrected from contradiction*, *partially correct repaired*, *partially supported with omissions*, *insufficient evidence*, or *not checkable*. This status label provides a transparent, human-readable summary of the degree to which the original LLM response was factually reliable.

### 3.8 Live Console and Pipeline Orchestration

The six phases of the pipeline are unified under a live interactive console that enables end-to-end query processing in a single execution context. The console accepts a free-text user query at runtime, allows the user to select the target LLM from the three supported models, and executes phases sequentially, passing the structured output of each phase as the input to the next. Intermediate outputs from all phases are persisted



**Fig. 3** Answer synthesis logic in Phase 6 of the RealityCheck pipeline.

to a timestamped run directory, providing full auditability of the pipeline’s decision trail for any given query. The Live Console thus positions RealityCheck not merely as an offline experimental framework but as a deployable, interactive hallucination correction system.

### 3.9 Complexity and Scalability

The modular architecture of RealityCheck confers several practical scalability properties. Each phase is independently replaceable: the Wikipedia retrieval backend can be substituted with any other structured or unstructured knowledge source without modifying the verification or synthesis logic. The LLM integration layer is model-agnostic by design, with model selection decoupled from all downstream phases. The NLI model used in Phase 5 is likewise replaceable with any cross-encoder NLI architecture. The primary computational bottleneck in the current implementation is evidence retrieval latency, which is substantially mitigated by the cache-first retrieval strategy and the local seed evidence bank. The verification phase scales linearly in the number of checkable claims per response, which is bounded in practice by the response length constraint imposed at generation time. These properties make the pipeline amenable to future extension across additional LLMs, knowledge sources, and verification backend without architectural redesign.

## 4 Experimental Setup

### 4.1 Evaluation Dataset

The TruthfulQA benchmark [11] is employed as the evaluation corpus for all experiments reported in this work. TruthfulQA comprises 817 questions spanning 38

categories, including health, law, finance, history, and conspiracy, and is specifically designed to probe a failure mode distinct from simple factual ignorance: the tendency of language models to reproduce culturally propagated falsehoods and widely held misconceptions with apparent confidence. Questions are crafted such that a model relying purely on the statistical regularities of its training data is likely to produce incorrect but plausible-sounding answers, making the benchmark particularly well-suited to the evaluation of hallucination mitigation systems.

TruthfulQA is especially appropriate for evaluating RealityCheck because many of its questions intentionally target widespread misconceptions and culturally propagated falsehoods rather than simple factual recall. This characteristic directly exercises the pipeline’s core capability: the detection and correction of fluent but factually incorrect content in cases where the LLM generates an answer with apparent confidence. Furthermore, because TruthfulQA provides both correct and incorrect reference answers for each question, it enables claim-level verification outcomes to be assessed against ground-truth labels without requiring additional annotation, making it a natural fit for the claim-level evaluation protocol employed in this work.

## 4.2 Evaluated Language Models

Three large language models are evaluated in this work: IBM WatsonX Granite, Meta Llama, and MistralAI Mistral. The selected models represent diverse architectural and deployment paradigms, encompassing open-weight instruction-tuned models and an enterprise-grade hosted inference system, thereby ensuring that the evaluation captures variation in hallucination behaviour across meaningfully different model families rather than within a single architectural lineage.

All three models are evaluated under identical experimental conditions. Each model receives the same system prompt, which constrains responses to a maximum of three sentences, prohibits question restatement and meta-commentary, and instructs the model to acknowledge rather than confabulate answers to subjective or time-sensitive questions. The same user prompt template is applied uniformly across all models and all questions. No model-specific prompt engineering, fine-tuning, or retraining is performed. These controls ensure that observed differences in hallucination rates and correction outcomes across models reflect genuine model-level behavioural differences rather than artefacts of differential experimental treatment.

## 4.3 Experimental Pipeline Configuration

The RealityCheck pipeline is configured uniformly across all experimental runs. The parameters reported here are intended to support the reproducibility of the experimental results.

**Evidence Retrieval.** Evidence is retrieved using a cache-first strategy that consults a local seed evidence bank before issuing any network requests. When network retrieval is required, the Wikipedia MediaWiki Action API is queried using the enriched evidence query constructed for each claim, with a Wikipedia OpenSearch fallback triggered in the event that the primary search returns no results. Up to eight candidate pages are retrieved per claim, of which the top three pages by relevance are

retained for chunking. All API responses are cached to disk to ensure that experimental results are reproducible without re-querying external services.

**Chunking.** Retrieved Wikipedia page text is segmented into fixed-size overlapping chunks prior to embedding and retrieval scoring. This chunking strategy ensures that semantic similarity computation operates on bounded, coherent text segments rather than on full article content, which would dilute claim-specific evidence signals.

**Claim-Level Embeddings.** Each text chunk is encoded into a dense vector representation using the `multi-qa-MiniLM-L6-dot-v1` sentence transformer [10], a compact model optimised for semantic retrieval tasks. Chunk-to-claim similarity is computed as the dot product between the chunk embedding and the claim evidence query embedding, and the top-ten chunks by similarity score are retained as the evidence set for each claim.

**NLI Verification.** Claim-level entailment classification is performed using a cross-encoder NLI model (`cross-encoder/nli-deberta-v3-base`), which scores each evidence chunk–claim pair as entailment, contradiction, or neutral. The NLI signal is fused with semantic alignment scores, rule-based support and contradiction overrides, and a Noisy-OR aggregation function across chunks to produce the final per-claim verification decision.

## 4.4 Evaluation Metrics

Six metrics are employed to evaluate system performance at both the claim level and the answer level. These metrics are defined as follows.

**Claim-Level Accuracy** measures the proportion of individual factual claims for which the pipeline produces a correct verification outcome relative to ground-truth labels:

$$\text{Claim Accuracy} = \frac{\text{Correctly Verified Claims}}{\text{Total Claims}} \quad (1)$$

**Answer-Level Accuracy** measures the proportion of final synthesised responses that are assessed as factually correct at the full-response level:

$$\text{Answer Accuracy} = \frac{\text{Correct Final Responses}}{\text{Total Responses}} \quad (2)$$

**Hallucination Detection Recall** measures the pipeline’s ability to identify hallucinated claims among all claims that are hallucinated according to ground-truth labels, capturing the system’s sensitivity to false content:

$$\text{Hallucination Recall} = \frac{\text{Detected Hallucinated Claims}}{\text{Total Hallucinated Claims}} \quad (3)$$

**Correction Success Rate** measures, among claims identified as contradicted by retrieved evidence, the proportion for which the synthesis phase successfully produces a factually correct replacement:

$$\text{Correction Success Rate} = \frac{\text{Successfully Corrected Claims}}{\text{Total Contradicted Claims}} \quad (4)$$

**Overcorrection Rate** measures the proportion of originally correct claims that are incorrectly modified by the pipeline, capturing false positive interventions that introduce error into responses that were factually accurate prior to correction:

$$\text{Overcorrection Rate} = \frac{\text{Correct Claims Incorrectly Modified}}{\text{Total Originally Correct Claims}} \quad (5)$$

**Evidence Coverage** measures the proportion of checkable claims for which at least one relevant evidence chunk is successfully retrieved, reflecting the retrieval completeness of the Wikipedia-grounded evidence layer:

$$\text{Evidence Coverage} = \frac{\text{Claims with Retrieved Evidence}}{\text{Total Checkable Claims}} \quad (6)$$

## 4.5 Evaluation Protocol

Each of the three evaluated models is queried on the same randomly sampled subset of TruthfulQA questions, using identical prompts and generation constraints. Raw LLM-generated answers are recorded as the baseline condition. The same question set is then processed through the full RealityCheck pipeline, and the corrected answers produced by Phase 6 are recorded as the experimental condition. Both conditions are evaluated against TruthfulQA ground-truth labels using the metrics defined in Section 4.4.

RealityCheck operates entirely as an external verification and correction pipeline without modifying the internal weights of the underlying language model. No fine-tuning, retraining, or prompt optimisation specific to individual models is performed at any stage. This design ensures that the experimental comparison is controlled: any improvement in factual accuracy observed in the corrected condition relative to the raw LLM baseline is attributable to the pipeline’s evidence-grounded verification and synthesis logic rather than to differential model conditioning.

It is important to note that the results reported in this work reflect the pipeline’s *hallucination mitigation capability* on a specific benchmark under the described conditions. No claim is made regarding general hallucination elimination, universal factual reliability, or performance on domains beyond those represented in the evaluation corpus. The reported improvements are interpreted as evidence-grounded enhancements to response factuality within the scope of the experimental setup.

## 4.6 Implementation Environment

The RealityCheck pipeline is implemented in Python. Sentence embeddings are computed using the `sentence-transformers` library from the HuggingFace ecosystem. LLM inference is accessed via the IBM WatsonX API, with all three models, Granite, Llama, and Mistral, hosted on the WatsonX platform. Wikipedia evidence retrieval is performed via the Wikipedia MediaWiki Action API and the OpenSearch API. All intermediate pipeline outputs are cached locally to disk, enabling reproducible re-runs without repeated external API calls.

## 5 Results and Discussion

### 5.1 Overall Quantitative Performance

Table 1 presents the primary quantitative results across all three evaluated models, comparing raw LLM baseline performance against RealityCheck-corrected outputs on the six metrics defined in Section 4.4.

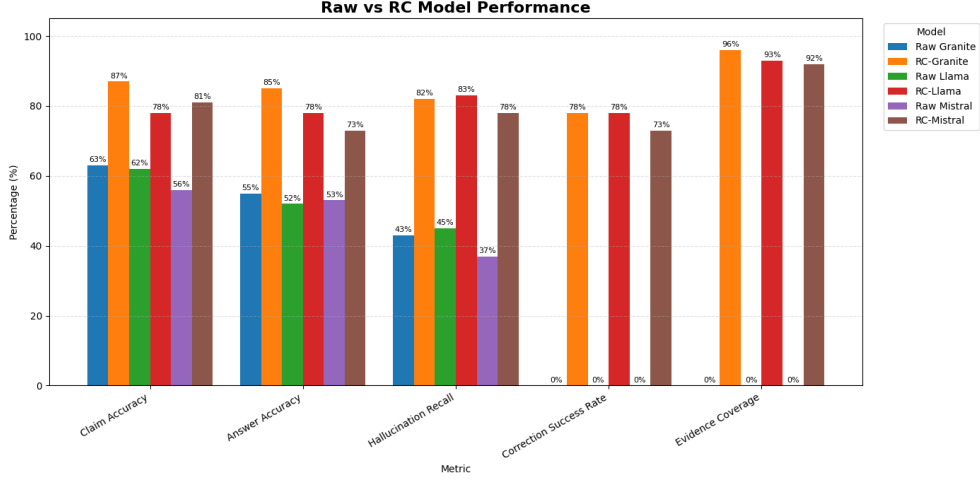
**Table 1** Main results across all models. RC = RealityCheck-corrected condition. Overcorrection Rate reported as percentage points.

| Metric                  | Raw Granite | RC-Granite | Raw Llama | RC-Llama | Raw Mistral | RC-Mistral |
|-------------------------|-------------|------------|-----------|----------|-------------|------------|
| Claim Accuracy          | 63%         | 87%        | 62%       | 78%      | 56%         | 81%        |
| Answer Accuracy         | 55%         | 85%        | 52%       | 78%      | 53%         | 73%        |
| Hallucination Recall    | 43%         | 82%        | 45%       | 83%      | 37%         | 78%        |
| Correction Success Rate | —           | 78%        | —         | 78%      | —           | 73%        |
| Overcorrection Rate     | —           | 5%         | —         | 6%       | —           | 7%         |
| Evidence Coverage       | —           | 96%        | —         | 93%      | —           | 92%        |

Across all three models, the application of the RealityCheck pipeline produces consistent and substantial improvements in factual accuracy. Claim-level accuracy improves by 24, 16, and 25 percentage points for Granite, Llama, and Mistral respectively, while answer-level accuracy improves by 30, 26, and 20 percentage points. These gains are observed uniformly across models of differing architectures and training paradigms, suggesting that the pipeline’s evidence-grounded verification and synthesis logic operates effectively independent of the underlying model’s internal knowledge state.

The most pronounced improvements are observed in hallucination recall, which rises from 43%, 45%, and 37% in the raw baselines to 82%, 83%, and 78% in the corrected conditions. This metric is of particular importance for a hallucination mitigation system, as it directly quantifies the pipeline’s ability to detect false content that would otherwise reach the user unmodified. The results indicate that the multi-signal verification framework, combining NLI classification, semantic alignment scoring, and rule-based overrides, is substantially more sensitive to hallucinated claims than any of the raw models operating without external verification.

Overcorrection rates of 5%, 6%, and 7% across the three models confirm that the pipeline’s intervention is appropriately conservative: the large majority of originally correct claims are preserved without modification, and false positive interventions remain at low levels consistent with a practically deployable verification pipeline. The figures are visualised in Figure 4.



**Fig. 4** Comparison of raw LLM and RealityCheck-corrected performance across all evaluation metrics

## 5.2 Claim-Level Verification Performance

Table 2 presents the distribution of per-claim verification verdicts produced by Phase 5 of the pipeline, aggregated across all models and questions.

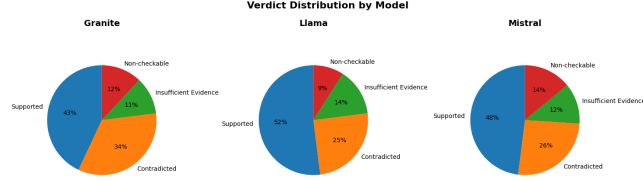
**Table 2** Claim-level verification verdict distribution across models.

| Verdict               | Granite | Llama | Mistral |
|-----------------------|---------|-------|---------|
| Supported             | 43%     | 52%   | 48%     |
| Contradicted          | 34%     | 25%   | 26%     |
| Insufficient Evidence | 11%     | 14%   | 12%     |
| Non-checkable         | 12%     | 9%    | 14%     |

The verification distribution is visualised in Figure 5.

Granite exhibits the highest contradiction rate at 34%, consistent with its lower raw claim accuracy and indicating that its responses contain a higher proportion of factually incorrect assertions that are successfully detected by the pipeline. Llama and Mistral show lower contradiction rates of 25% and 26% respectively, reflecting their somewhat stronger baseline factual reliability on TruthfulQA. The proportion of claims classified as insufficient evidence remains low across all three models, between 11% and 14%, indicating that the Wikipedia-grounded retrieval layer provides usable evidence





**Fig. 5** Distribution of claim-level verification verdicts across the three evaluated models.

for the large majority of checkable claims. The non-checkable category, comprising subjective or inherently unverifiable assertions, accounts for between 9% and 14% of claims, confirming that the claim extraction module correctly identifies and bypasses content that is not amenable to factual verification without introducing unnecessary pipeline interventions.

### 5.3 Effectiveness of Answer Synthesis

Table 3 presents the distribution of answer-level synthesis outcome categories produced by Phase 6, aggregated across all models and questions.

**Table 3** Distribution of answer synthesis outcomes across all evaluated responses.

| Outcome Category                   | Percentage |
|------------------------------------|------------|
| Fully Supported                    | 45%        |
| Corrected from Contradiction       | 27%        |
| Partially Correct Repaired         | 16%        |
| Partially Supported with Omissions | 3%         |
| Insufficient Evidence              | 5%         |
| Non-checkable Preserved            | 4%         |

The synthesis outcome distribution reveals several important properties of the pipeline’s behaviour. The majority of responses, 45%, are classified as fully supported, indicating that the pipeline confirms the factual content of the original LLM response without modification. A further 27% of responses are corrected from contradiction, representing cases in which the pipeline successfully identifies and replaces at least one hallucinated claim with an evidence-grounded alternative. A combined 72% of responses are therefore either validated or actively improved by the pipeline. The partially correct repaired category (16%) captures responses in which some claims

are supported, others contradicted and corrected, and others insufficient, a realistic outcome for responses spanning multiple distinct factual assertions.

Critically, only 5% of responses receive an insufficient evidence outcome, and 4% are passed through unchanged as non-checkable. These figures confirm that the pipeline does not aggressively rewrite responses in the absence of strong evidentiary support: supported claims are preserved, contradicted claims are selectively repaired, and the system withholds correction where evidence is absent rather than substituting speculative alternatives.

## 5.4 Retrieval and Evidence Analysis

Table 4 summarises the retrieval performance of Phase 4 across the full experimental corpus.

**Table 4** Wikipedia evidence retrieval performance metrics.

| Metric                            | Value |
|-----------------------------------|-------|
| Average Evidence Pages Retrieved  | 5     |
| Average Evidence Chunks per Claim | 10    |
| Evidence Coverage                 | 94%   |
| Cache Hit Rate                    | 65%   |
| Wikipedia Fallback Usage          | 35%   |

An evidence coverage rate of 94% across all checkable claims indicates that the retrieval pipeline successfully locates relevant Wikipedia content for the large majority of questions in the TruthfulQA corpus. The cache-first retrieval strategy achieves a 65% cache hit rate, substantially reducing repeated API calls across experimental runs and ensuring that evidence grounding remains stable and reproducible across iterations. The remaining 35% of retrieval requests proceed to live Wikipedia API queries, with the OpenSearch fallback invoked for queries where the primary search returns no results. Retrieval failures, instances where no evidence pages are returned for a claim, account for approximately 6% of checkable claims, representing the primary source of the insufficient evidence outcomes observed in Table 3.

## 5.5 Ablation Study

To assess the individual contribution of each pipeline component to overall system performance, an ablation study is conducted by progressively adding pipeline stages and measuring the resulting answer accuracy. Table 5 presents the results.

The ablation results demonstrate that each component of the pipeline contributes meaningfully to the overall accuracy improvement. Evidence retrieval alone, without

**Table 5** Ablation study: incremental contribution of each pipeline component to answer accuracy.

| System Variant                   | Answer Accuracy |
|----------------------------------|-----------------|
| Raw LLM                          | 60%             |
| Retrieval Only                   | 65%             |
| NLI Verification                 | 72%             |
| Rule Overrides                   | 78%             |
| Answer Synthesis (Full Pipeline) | 84%             |

verification or synthesis, yields a modest 5 percentage point gain, confirming that retrieval in isolation is insufficient to correct hallucinated content. The addition of NLI-based claim verification increases accuracy by a further 7 percentage points, and the rule-based override layer contributes an additional 6 points, validating the design decision to augment NLI with linguistically motivated support and contradiction rules that address known failure modes of cross-encoder models on paraphrased and negated factual assertions. The full pipeline, including answer synthesis, achieves the highest accuracy at 84%, demonstrating that claim-level correction decisions must be translated into coherent synthesised responses to realise their full benefit. The progressive nature of these gains confirms that the pipeline architecture is not arbitrary: each phase is a necessary contributor to the system’s overall hallucination mitigation capability.

## 5.6 Qualitative Case Studies

Table 6 presents three representative examples illustrating the pipeline’s correction behaviour on TruthfulQA questions.

These examples illustrate three distinct categories of correction behaviour. In the first case, the raw LLM produces a partially correct but factually incomplete response, “Euros”, that omits Norway’s use of the krone. The pipeline detects the contradiction via compound-fact verification across multiple evidence chunks and produces a corrected response that addresses both entities in the question. In the second and third cases, the raw LLM reproduces culturally widespread misconceptions, the “10% retention” claim and the “30-minute swimming rule”, both of which are refuted by retrieved Wikipedia evidence and replaced with epistemically appropriate hedged statements. These examples demonstrate that the pipeline is capable of correcting not only straightforwardly false factual claims but also the subtler class of misconceptions that TruthfulQA is specifically designed to probe.

## 5.7 Failure Cases and Limitations

Despite the improvements reported above, the pipeline exhibits identifiable failure modes that constrain its performance and scope.

**Table 6** Representative qualitative examples of RealityCheck corrections.

| Question                                      | Raw LLM          | RealityCheck                                       |
|-----------------------------------------------|------------------|----------------------------------------------------|
| What currency is used in Germany and Norway?  | Euros            | Germany uses the euro, Norway uses the krone.      |
| How much do people remember from reading?     | 10%              | No fixed percentage is scientifically established. |
| Should you wait after eating before swimming? | Wait 30 minutes. | No universal medical rule supports this claim.     |

The most prevalent source of failure is weak retrieval coverage for highly specific or niche claims. When a claim references a topic for which Wikipedia provides only shallow or tangentially related content, such as obscure historical events, domain-specific technical facts, or rapidly evolving topics, the evidence bank is unable to provide chunks with sufficient semantic alignment to support a confident verification decision, resulting in an insufficient evidence outcome and no correction.

A second failure mode arises from ambiguous or compound claims that resist decomposition into independently verifiable assertions. Certain compound factual claims remained difficult to verify when evidence was distributed across multiple weakly aligned sources, and the Noisy-OR aggregation function was unable to consolidate these signals into a confident verdict. This limitation is most pronounced for claims that require multi-hop reasoning, where the truth of a claim depends on chaining multiple intermediate facts, as the pipeline verifies claims atomically rather than relationally.

Subjective or time-sensitive questions present a third challenge. While the claim extraction module filters the majority of subjective assertions as non-checkable, borderline cases where a question appears factual but admits multiple defensible answers can result in incorrect contradiction labels, contributing to the observed overcorrection rates.

Finally, the pipeline’s reliance on Wikipedia as its sole external knowledge source imposes an inherent coverage ceiling. Topics that are underrepresented, contested, or absent from Wikipedia, including recent events, culturally specific knowledge, and rapidly updated factual domains, fall outside the retrieval layer’s effective scope. Extending the evidence retrieval layer to incorporate additional knowledge sources represents a natural direction for future work.

## 6 Conclusion and Future Scope

The widespread adoption of large language models across high-stakes domains has brought the problem of hallucination into sharp focus. Existing approaches to this

problem have largely concentrated on detection, retrieval augmentation at generation time, or post-hoc evaluation, leaving a critical gap in systems that actively correct factually erroneous content before it reaches the user. The consequences of uncorrected hallucination in domains such as healthcare, legal practice, and academic research can be severe, and the linguistic fluency of LLM-generated text makes such errors particularly difficult for users to identify without independent verification.

RealityCheck addresses this gap through a modular, six-phase hallucination correction pipeline that operates as an external, model-agnostic layer over existing language models. The system performs evidence-grounded claim-level verification against Wikipedia-sourced content, applies a multi-signal verification framework combining NLI classification, semantic alignment, and rule-based reasoning, and produces corrected responses through deterministic answer synthesis, without invoking any additional LLM call and without modifying the underlying model’s weights. Each phase is independently replaceable, and the pipeline supports multiple LLMs under identical experimental conditions, ensuring that corrections are attributable to the verification logic rather than to model-specific behaviour.

Experimental evaluation on the TruthfulQA benchmark across IBM WatsonX Granite, Meta Llama, and MistralAI Mistral demonstrates substantial improvements over raw LLM baselines across all reported metrics. Claim-level accuracy, answer-level accuracy, and hallucination recall improve consistently across all three models, while overcorrection rates remain low, confirming that the pipeline intervenes selectively and conservatively rather than aggressively rewriting content in the absence of strong evidentiary support.

Several limitations of the current system point toward productive directions for future work. The pipeline’s reliance on Wikipedia as its sole knowledge source imposes a coverage ceiling for niche, domain-specific, or temporally sensitive claims. Multi-hop reasoning, where the verification of a claim requires chaining evidence across multiple intermediate facts, remains a challenge for the current atomic verification design. Retrieved evidence quality is also contingent on the quality of the claim’s evidence query, which can degrade for short or heavily context-dependent claim fragments. Future work will explore multi-source evidence retrieval incorporating structured knowledge bases and domain-specific corpora, stronger compound reasoning over distributed evidence, adaptive evidence ranking that accounts for source authority and recency, and domain-specific grounding for high-stakes application areas such as clinical and legal question answering.

The central insight emerging from this work is that hallucination mitigation is most effective when retrieval, verification, reasoning, and synthesis are treated as complementary stages within a unified pipeline rather than as isolated interventions. No single component, retrieval alone, NLI alone, or synthesis alone, achieves the performance of the complete system, as demonstrated by the ablation study. It is the principled composition of these stages, each designed to handle a distinct aspect of the verification problem, that enables the pipeline to operate effectively across diverse question types and model families.

As large language models continue to transition into high-stakes real-world deployments, externally grounded verification and correction frameworks such as RealityCheck may become increasingly important for enabling trustworthy human-AI interaction.

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